

Numerical Verification Of Hagen-Poiseuille Flow In Rectangular Microchannel



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Introduction

Hagen–Poiseuille flow describes steady, laminar, pressure-driven motion of an incompressible Newtonian fluid through long ducts or channels.

For fully developed internal viscous flow, the **Navier–Stokes equations simplify** to a 2-D Poisson equation in the transverse plane.

Unlike circular pipes, **rectangular microchannels lack radial symmetry**, resulting in:

- Non-parabolic velocity distribution
- Low-velocity corner regions
- Strong dependence on channel geometry

The velocity field in rectangular ducts is obtained analytically using **separation of variables**, resulting in an infinite Fourier series solution.

Literature Review

Internal viscous flow in microchannels is governed by the Navier–Stokes equations for incompressible Newtonian fluids.

Under assumptions of:

- Laminar flow
- Steady state
- Fully developed velocity field the governing equations reduce to a **2-D Poisson equation** in the transverse directions.

In rectangular ducts:

- Velocity profile becomes non-parabolic
- Corner regions influence momentum transport
- Analytical solution requires Fourier series expansion

Literature Review

The **Darcy–Weisbach equation** is used to evaluate friction factor by relating:

- Pressure drop
- Hydraulic diameter
- Reynolds number
- Mean velocity

Literature indicates that applying **circular pipe correlations** to rectangular microchannels may result in:

- Significant pressure drop prediction errors
- Incorrect estimation of energy losses

Motivation and objectives

Motivation

Microchannels are widely used in microfluidic applications such as lab-on-chip devices, drug delivery systems, and biomedical diagnostics, where accurate prediction of pressure drop and flow behavior is essential for reliable design. Numerical verification of Hagen–Poiseuille flow with analytical solutions helps validate computational tools like MATLAB and ANSYS Fluent for rectangular microchannels, where hydraulic resistance depends on aspect ratio but remains independent of flow rate in laminar flow. This study bridges the gap between theoretical analysis and numerical simulation in microscale flow systems where experimental validation is difficult.

Objective

- To numerically verify **Hagen-Poiseuille flow characteristics** in a rectangular microchannel using CFD.
- To analytically compute pressure drop using MATLAB for fully developed laminar flow conditions.
- To demonstrate that **hydraulic resistance is independent of flow rate** for laminar incompressible flow.
- To study the effect of **channel aspect ratio** on pressure drop and average velocity

Methodology

The numerical verification of laminar Hagen–Poiseuille flow in a rectangular microchannel was carried out using both analytical and computational approaches.

The study involved:

- Modeling pressure-driven internal viscous flow
- Evaluating pressure drop across the microchannel
- Determining hydraulic resistance and friction factor
- Performing comparative analysis between theoretical and simulated results

Two independent approaches were used:

- Analytical solution using MATLAB
- Numerical simulation using ANSYS Fluent

The results obtained from CFD were validated against the analytical solution.

Methodology

To analyze the flow characteristics in rectangular microchannels, two separate cases were considered:

Case 1: Varying Flow Rate

- Channel geometry kept constant
- Flow rate varied
- Pressure drop evaluated
- Hydraulic resistance calculated

To verify independence of hydraulic resistance from flow rate in laminar regime.

Case 2: Varying Aspect Ratio

- Mass flow rate kept constant
- Channel aspect ratio varied
- Pressure drop and average velocity calculated
- Friction factor evaluated

To study the influence of geometry on flow resistance.

Methodology

Numerical Approach (Ansys Fluent)

1. Geometry and Domain

A 3D rectangular microchannel was created with the following dimensions:

- **Height (H)** = 200 μm (Height varied for case 2)
- **Width (W)** = 200 μm
- **Length (L)** = 2 mm

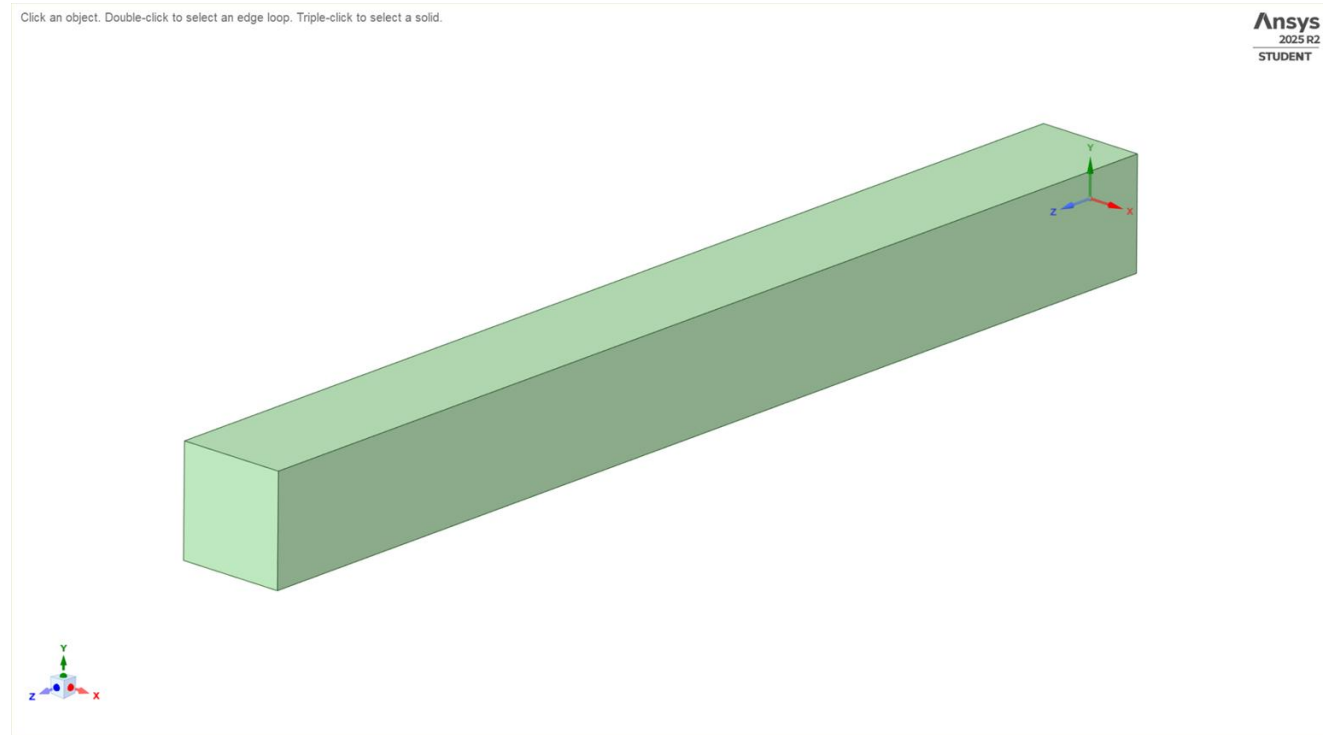


Fig 1: Rectangular microchannel

Methodology

2. Meshing

- Structured mesh generated over the computational domain.
- Fine mesh resolution ensured numerical stability and accuracy in pressure drop prediction.
- Mesh biasing was given to ensure accurately capture velocity gradients at boundary.

	No of divisions on edge			
	40	50	60	70
Pressure drop(Pa)	6.4816407	6.5269946	6.5630728	6.5858068
Avg Velocity at outlet (m/s)	0.0041745	0.0041747	0.0041747	0.0041748

Table 1: Grid independence test

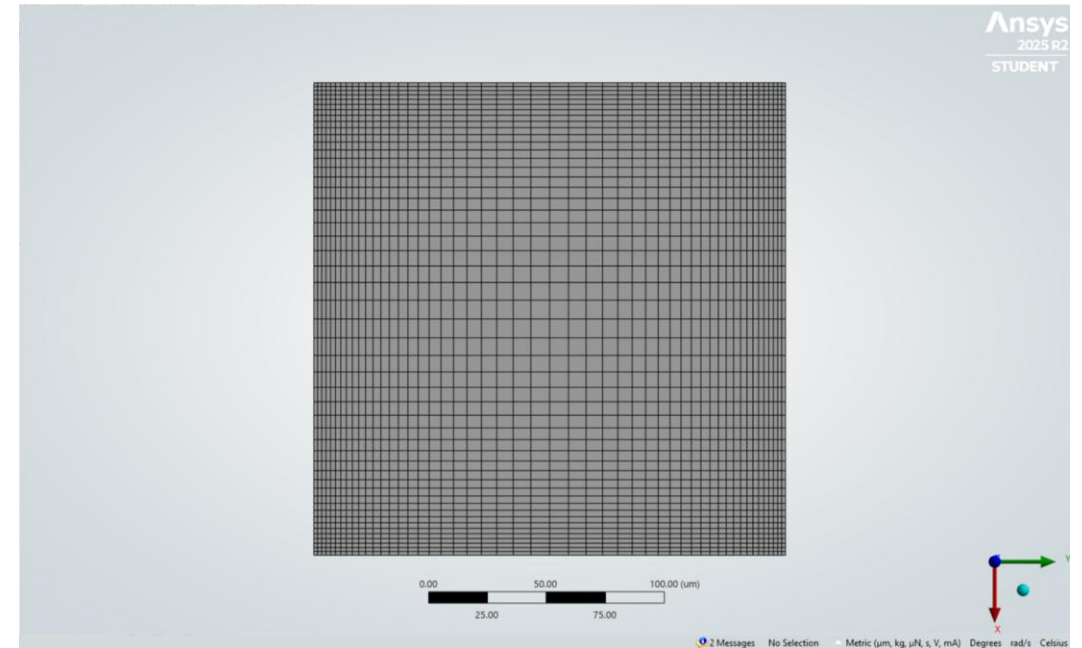


Fig 2: Mesh division and biasing

Methodology

3. Solver setup

Solver Type : Pressure-Based Solver

Time : Steady-State Simulation

Viscous Model : Laminar

Material Properties

Working Fluid : Water (at 300 K)

- Density (ρ) = **988 kg/m³**
- Dynamic Viscosity (μ) = **0.001 Pa·s**

4. Boundary Conditions

Inlet

- **Type** : Mass Flow Inlet
- **Mass Flow Rate** : 1.667E-07 kg/s to 6.667E-07 kg/s
- Inlet mass flow rate was varied according to simulation cases.

Outlet

- **Type** : Pressure Outlet
- **Gauge Pressure** : 0 Pa

Walls

- **Wall Condition** : Stationary Walls
- **No-Slip Boundary Condition Applied**

Methodology

Analytical Approach (MATLAB)

Fundamental Equation: The velocity field $u(y, z)$ is solved using a Fourier series to satisfy the no-slip boundary conditions on all four walls.

$$u(y, z) = \frac{16a^2}{\pi^3} \left(\frac{-dP/dx}{\mu} \right) \sum_{k=1}^{\infty} \frac{(-1)^{(n-1)/2}}{n^3} \left[1 - \frac{\cosh(n\pi z/2a)}{\cosh(n\pi b/2a)} \right] \cos \left(\frac{n\pi y}{2a} \right)$$

Geometric Adaptability: By varying the Aspect Ratio ($AR = H/W$), the code transitions from a square "bullet" profile to a 2D "flat" profile.

Implementation: Your MATLAB script computes 100 terms ($n=100$) to ensure series convergence, resulting in an integration error of only **0.001%**.

Methodology

- Hydraulic resistance was calculated using:

$$R_h = \frac{12\mu L}{wh^3 \left[1 - 0.63 \left(\frac{h}{w}\right)\right]}$$

- Pressure drop was obtained from:

$$\Delta P = R_h \dot{Q}$$

- Friction factor was determined using the **Darcy–Weisbach equation**

$$f = \frac{\Delta P}{\left(\frac{L}{D_h}\right) \left(\frac{\rho \cdot u_{mean}^2}{2}\right)}$$

Results and discussion

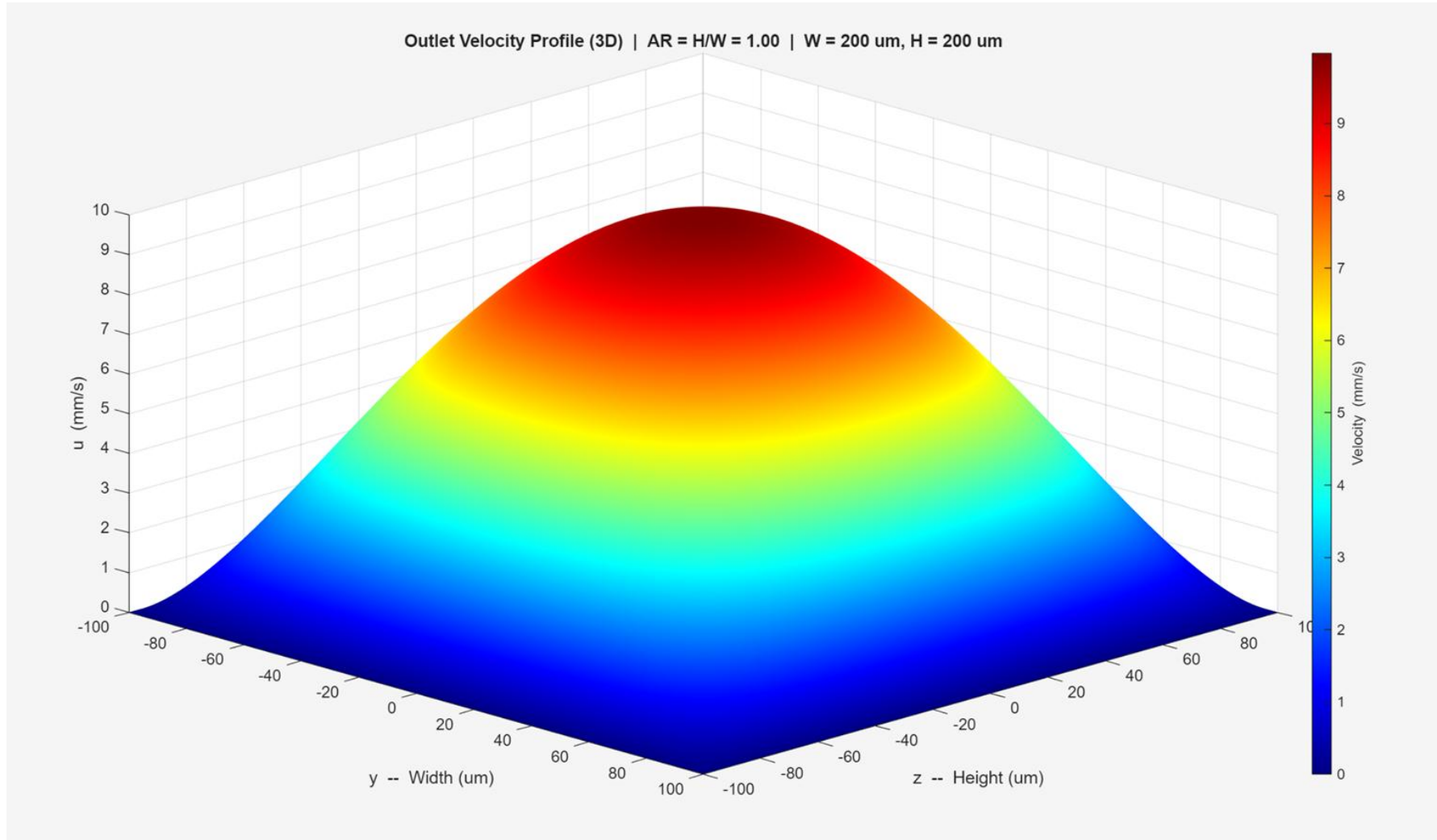


Fig. 3 3D velocity profile at outlet

Results and discussion

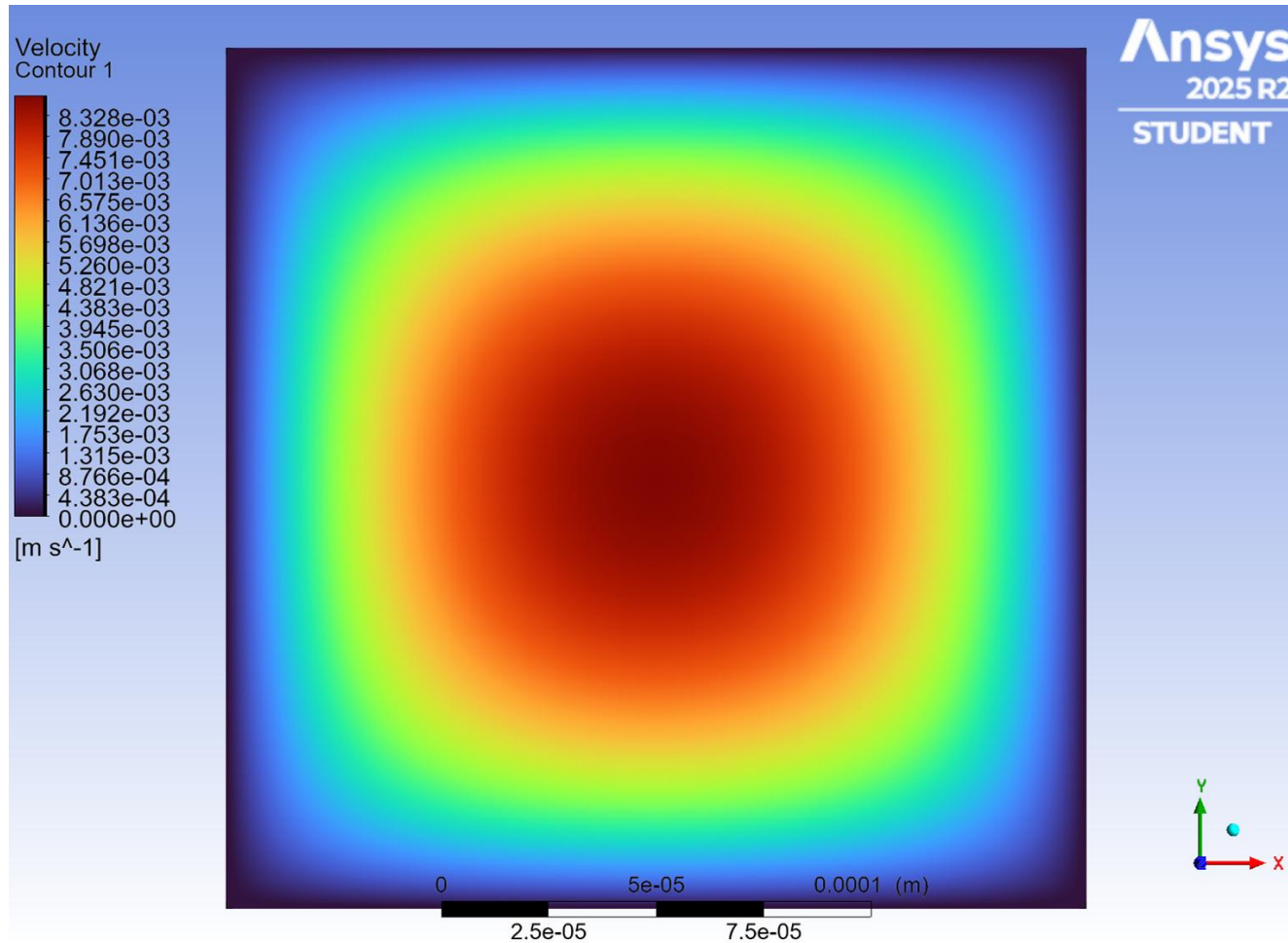


Fig. a Numerical result

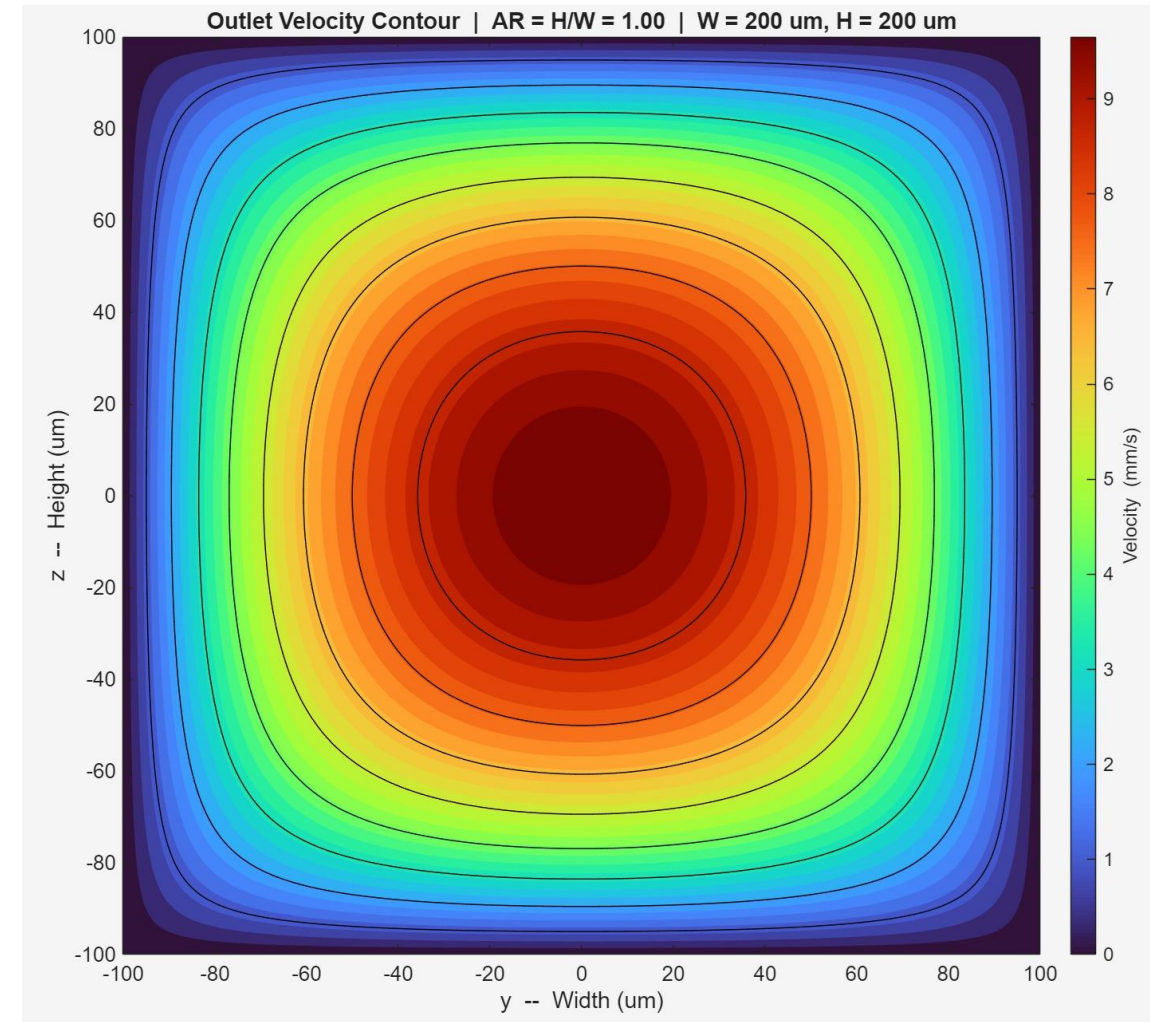


Fig. b Analytical result

Fig. 4 Velocity profile at outlet for AR 1

Results and discussion

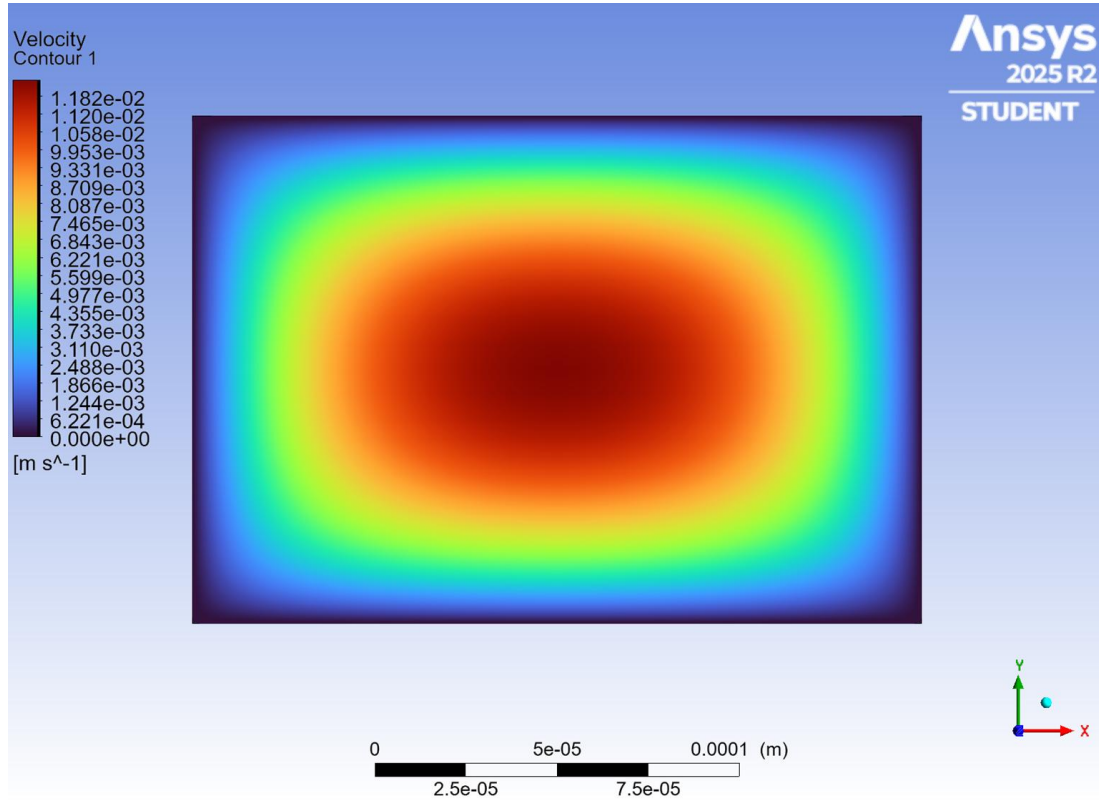


Fig. a Numerical result

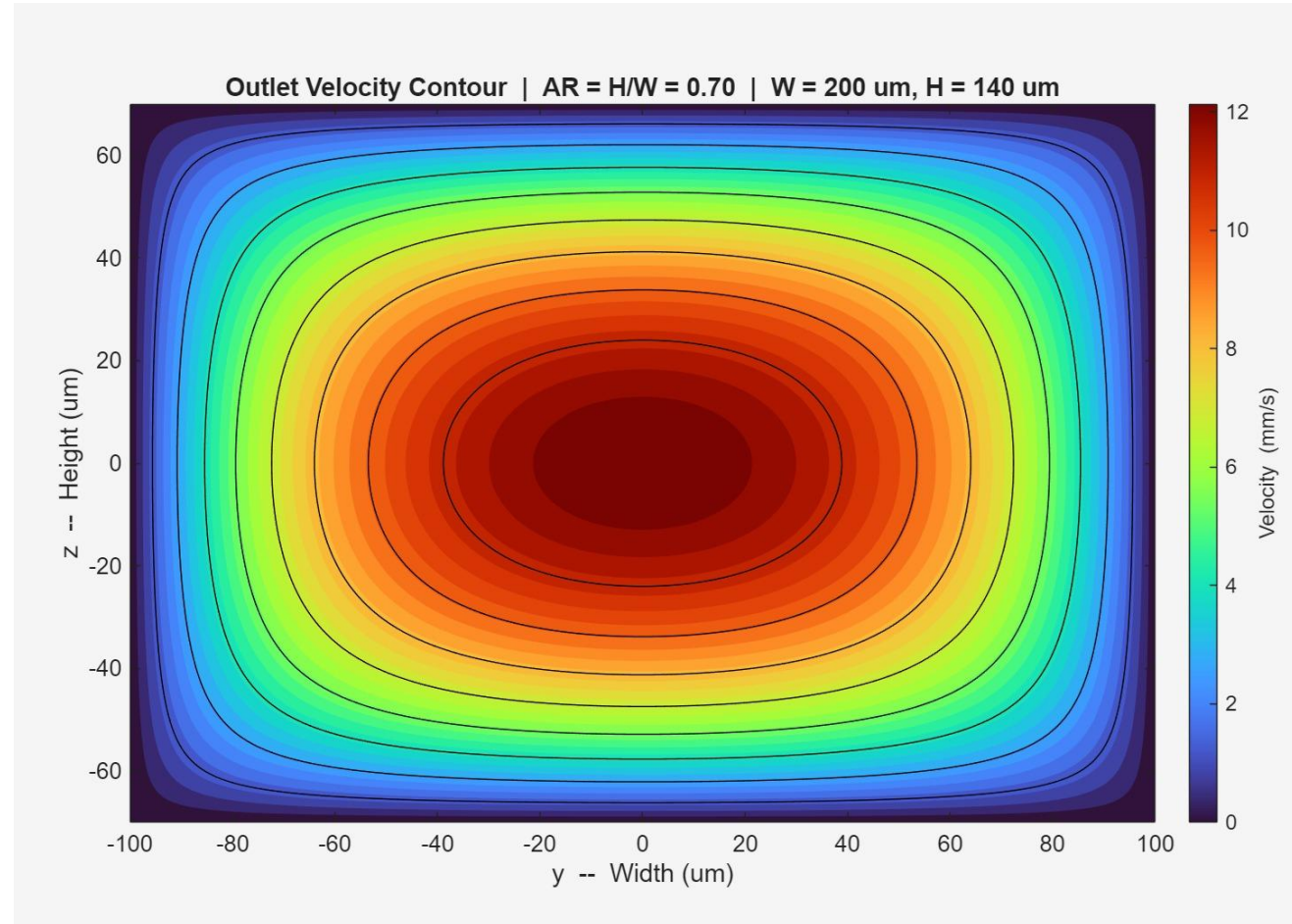


Fig. b Analytical result

Fig. 5 Velocity profile at outlet for AR 0.7

Results and discussion

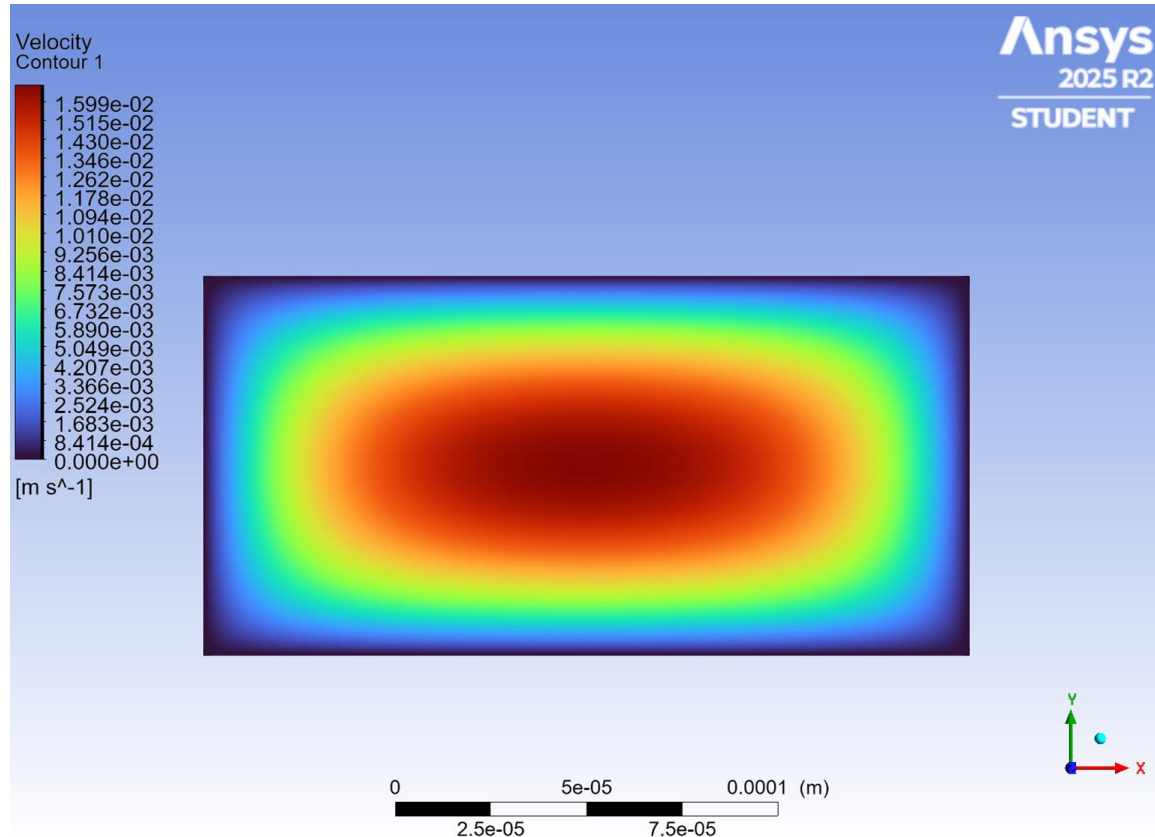


Fig. a Numerical result

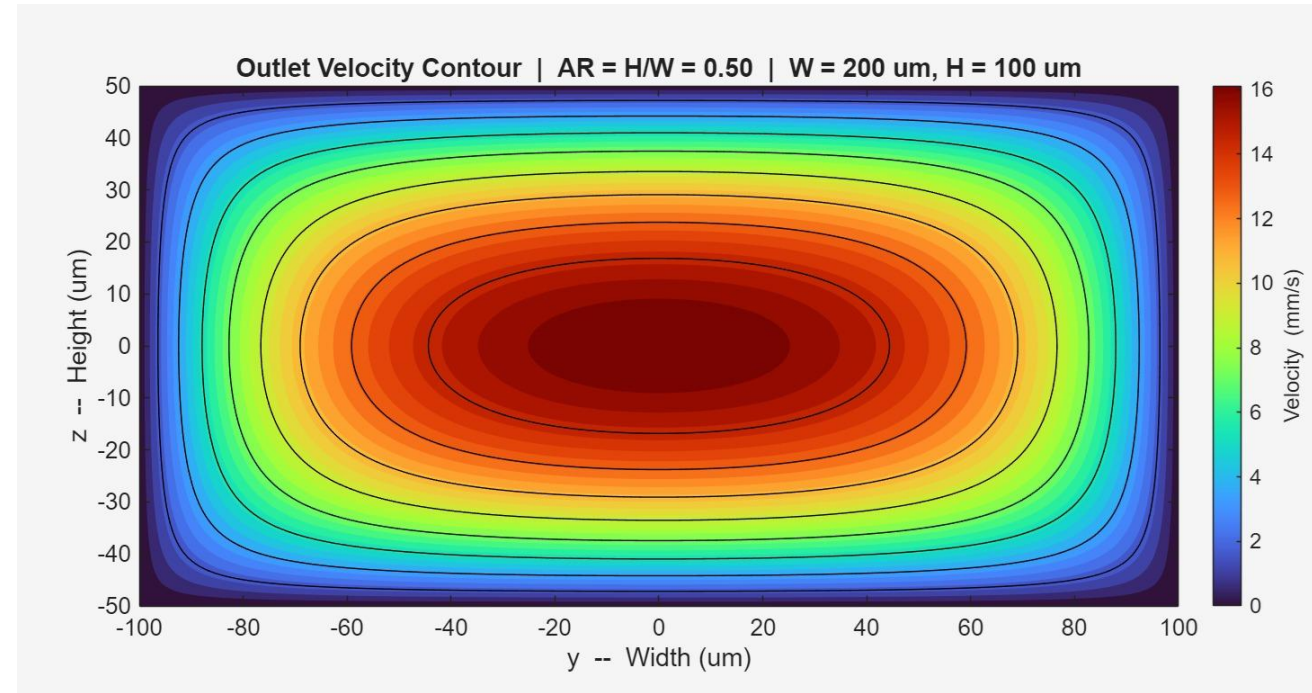


Fig. b Analytical result

Fig. 6 Velocity profile at outlet for AR 0.5

Results and discussion

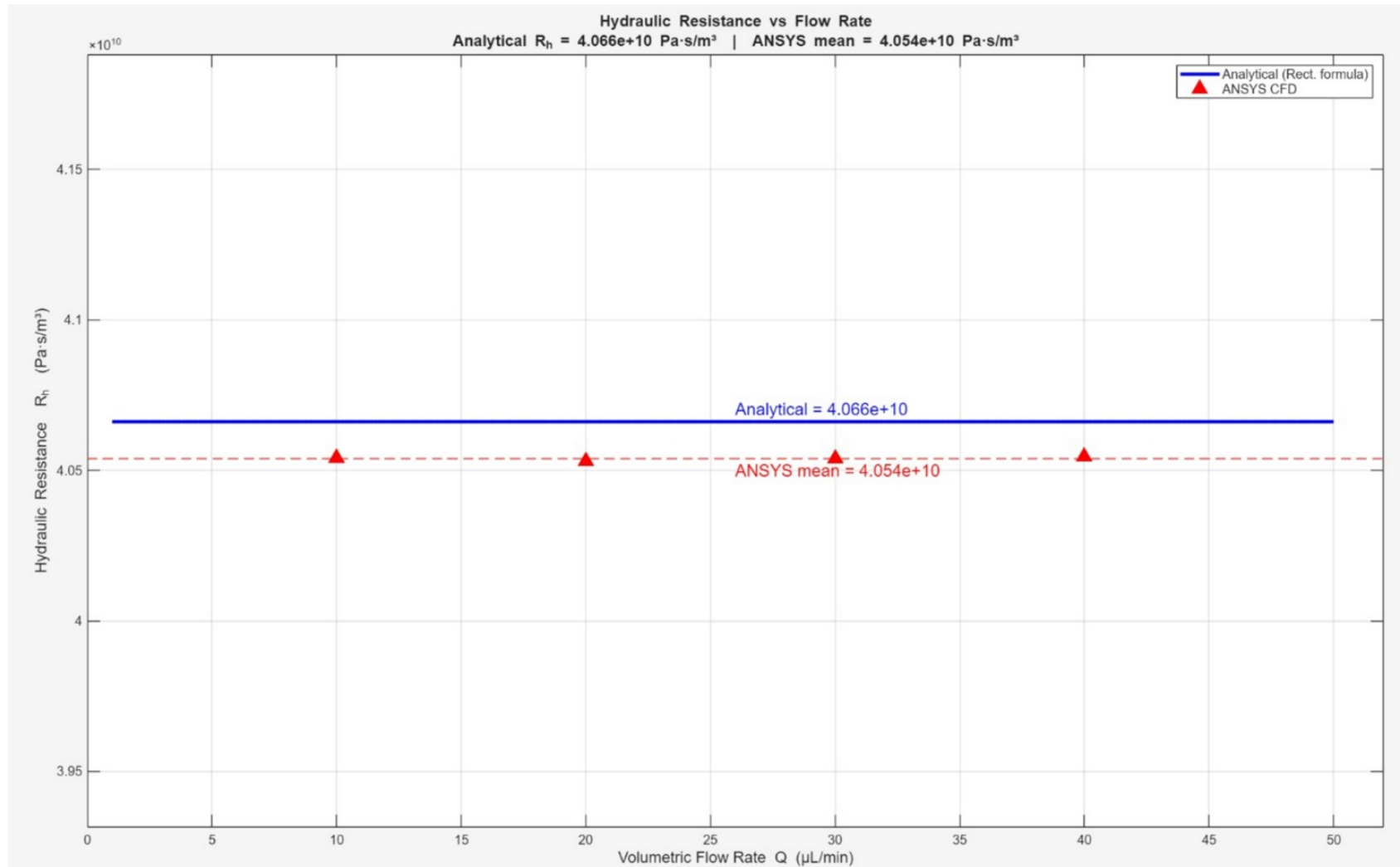


Fig. 7 Hydraulic resistance v/s Flow rate

Results and discussion

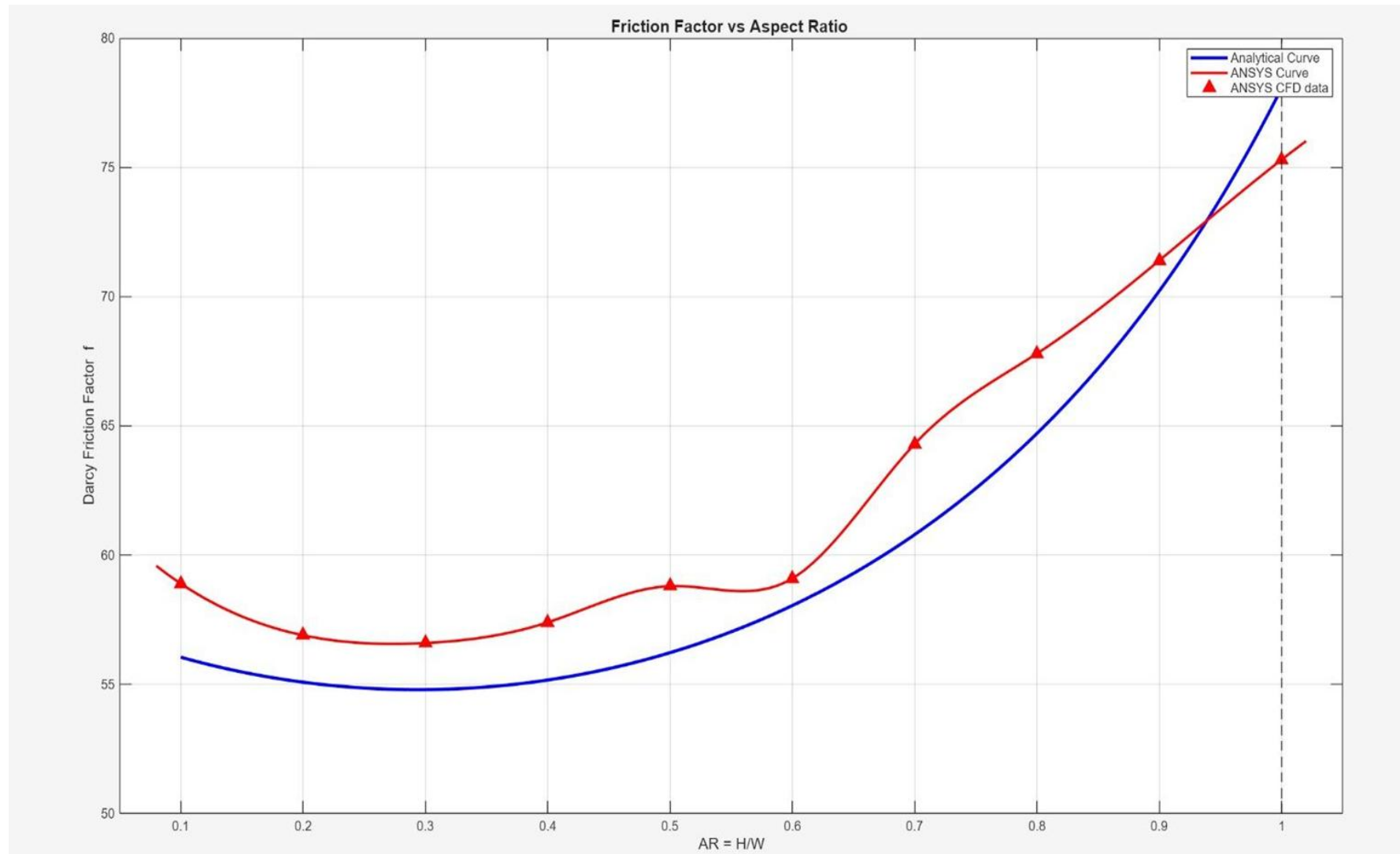


Fig. 8 Friction Factor v/s Aspect Ratio

Conclusions

- Laminar Hagen–Poiseuille flow in a rectangular microchannel was successfully simulated using **MATLAB (analytical)** and **ANSYS Fluent (numerical)**.
- The CFD results showed good agreement with the analytical solution, validating the numerical model.
- For varying volumetric flow rate:
 - Pressure drop increased linearly with flow rate.
 - Hydraulic resistance remained constant for fixed geometry, confirming theoretical predictions.
- For varying aspect ratio:
 - Friction factor strongly depended on channel geometry.
 - Decreasing aspect ratio (narrower channel height) significantly increased flow resistance.
 - Corner effects in rectangular ducts influenced the velocity distribution and pressure drop.
- The study confirms that:
 - Analytical models are reliable for laminar microchannel flow.
 - CFD tools like ANSYS Fluent accurately capture rectangular duct behavior.
 - Channel geometry plays a critical role in microfluidic design.

Future scope

- **Extension to Transient Flow Conditions**
In this work, steady-state flow is considered. Future studies can include time-dependent (unsteady) flow to analyze micro-pump driven systems and pulsatile flow conditions.
- **Study of Non-Newtonian Fluids**
The present analysis assumes a Newtonian fluid. Many real microfluidic applications involve bio-fluids or polymer solutions, which show non-Newtonian behavior. Extending the model to such fluids will make the study more practical.
- **Slip Flow and Rarefaction Effects**
At very small scales, the no-slip boundary condition may not hold perfectly. Future work can incorporate slip boundary conditions and analyze flows at higher Knudsen numbers.
- **Thermal Coupled Analysis**
The current study focuses only on hydrodynamics. In microchannel cooling applications, heat transfer plays a major role. A coupled fluid–thermal simulation can provide deeper insights.
- **Effect of Surface Roughness**
Real microchannels are not ideally smooth. Including wall roughness in numerical simulations can improve prediction accuracy of friction factor and pressure drop.

References

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